

Grade 6 Accelerated
Unit 13
Lesson 11 Homework

Name Answer Key
Date _____
Period _____

From the specified set of integers, determine which, if any, make the equation true.

$\{-4, -1, 1, 8\}$

- | | |
|---|--|
| <p>1. $12 + 5w = 6w + 4$
$12 + 5(-4) = 6(-4) + 4$
$12 - 20 = -24 + 4$
$-8 \neq -20$ NOT TRUE</p> <p>$12 + 5(-1) = 6(-1) + 4$
$12 - 5 = -6 + 4$
$7 \neq -2$ NOT TRUE</p> <p>$12 + 5(1) = 6(1) + 4$
$12 + 5 = 6 + 4$
$17 \neq 10$ NOT TRUE</p> <p>$12 + 5(8) = 6(8) + 4$
$12 + 40 = 48 + 4$
$52 = 52$ TRUE</p> | <p>2. $3 - x = 2x + 6$
$3 - (-4) = 2(-4) + 6$
$3 + 4 = -8 + 6$
$7 \neq -2$ NOT TRUE</p> <p>$3 - (-1) = 2(-1) + 6$
$3 + 1 = -2 + 6$
$4 = 4$ TRUE</p> <p>$3 - (1) = 2(1) + 6$
$3 - 1 = 2 + 6$
$2 \neq 8$ NOT TRUE</p> <p>$3 - (8) = 2(8) + 6$
$3 - 8 = 16 + 6$
$-5 \neq 22$ NOT TRUE</p> |
| <p>3. $3y - 5 = 8y + 15$
$3(-4) - 5 = 8(-4) + 15$
$-12 - 5 = -32 + 15$
$-17 = -17$ TRUE</p> <p>$3(-1) - 5 = 8(-1) + 15$
$-3 - 5 = -8 + 15$
$-29 \neq -49$ NOT TRUE</p> <p>$3(1) - 5 = 8(1) + 15$
$3 - 5 = 8 + 15$
$-2 \neq 23$ NOT TRUE</p> <p>$3(8) - 5 = 8(8) + 15$
$24 - 5 = 64 + 15$
$19 \neq 79$ NOT TRUE</p> | <p>4. $2u + 6 = 9u - 1$
$2(-4) + 6 = 9(-4) - 1$
$-8 + 6 = -36 - 1$
$-2 \neq -37$ NOT TRUE</p> <p>$2(-1) + 6 = 9(-1) - 1$
$-2 + 6 = -9 - 1$
$4 \neq -10$ NOT TRUE</p> <p>$2(1) + 6 = 9(1) - 1$
$2 + 6 = 9 - 1$
$8 = 8$ TRUE</p> <p>$2(8) + 6 = 9(8) - 1$
$16 + 6 = 72 - 1$
$22 \neq 71$ NOT TRUE</p> |

Determine if the inequality $18 \geq 2m - 6$ is true for each value.

5. 8

$$18 \geq 2(8) - 6$$

$$18 \geq 16 - 6$$

$$18 \geq 10$$

True

6. 12

$$18 \geq 2(12) - 6$$

$$18 \geq 24 - 6$$

$$18 \geq 18$$

True

7. 15

$$18 \geq 2(15) - 6$$

$$18 \geq 30 - 6$$

$$18 \geq 24$$

False

Determine if the inequality $-y + 3 + 8y \geq -3y - 7$ is true for each value.

8. -5

$$-(-5) + 3 + 8(-5) \geq -3(-5) - 7$$

$$5 + 3 - 40 \geq 15 - 7$$

$$-32 \geq 8$$

False

9. 0

$$3 + 7y \geq -3y - 7$$

$$3 + 7(0) \geq -3(0) - 7$$

$$3 \geq -7$$

True

10. 5

$$3 + 7y \geq -3y - 7$$

$$3 + 7(5) \geq -3(5) - 7$$

$$3 + 35 \geq -15 - 7$$

$$38 \geq -22$$

True